

Tonal Contrasts in the Malipo Variety of the Mienic Language

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Abstract

This paper describes tonal contrasts in the Malipo variety of the Mienic language and explores how they are manifested by acoustic cues. Results show that Linear Mixed-Effects Models based on Growth Curve Analysis successfully characterizes the 5 level tones, 3 falling tones, 3 rising or concave tones, and 2 convex tones in the Malipo variety. In addition to F0, duration also plays a role in a complex tonal system.

Index Terms: tone, pitch, contour, duration, Mienic language

1. Introduction

The Hmong-Mien languages are known for their complexity in tones, with tones primarily serving to distinguish lexical meanings [1, pp. 11]. There are four major varieties or dialects in the Mienic language family, namely Mien, Kim Mun, Biao Min, and Zao Min. And the Kim Mun dialects have the richest inventories of tones, typically 12 to 15 tones, featuring an extremely complex tonal system with multiple layers of tonal split [2, pp. 158-162]. This paper describes tonal contrasts in the Malipo variety and explores how they are manifested by acoustic cues.

Table 1: *Tones in the Malipo variety.*

	Tone	Item	Gloss
unchecked	T1	pu213	three
	T1'&T7'a	pu51	to go prone
	T2	du21	nine
	T3	pu341	to burn
	T3'	bu33	blind
	T4	pu31	hand
	T5	bu35	name
	T5'	pu351	wealthy
checked	T6	pu22	to incubate
	T7a	da:t212	wing
	T7b	dat33	to weave
	T7'b	dat44	to wash (rice)
	T8	tap22	bean
		ta:p22	to wear (shoes)

The Malipo variety belongs to the Kim Mun dialects, and is spoken by the Landian Yao people in Malipo County and its vicinity in the Wenshan Zhuang and Miao Autonomous Prefecture of Yunnan Province, China. The Malipo variety has 13 tones. As shown in Table 1, the 9 tones in unchecked syllables are denoted as T1, T1'&T7'a (two merged tones), T2, T3, T3', T4, T5, T5' and T6, and the 4 tones in checked syllables with a stop ending are labeled as T7a, T7b, T7'b and

T8. The tones are numbered according to the comparative tonal phonology in the Mienic languages so that one can examine tonal development in the Malipo variety. First, odd-numbered tones are associated with historical voiceless initials, and even-numbered tones with historical voiced initials [3, 4, 5]. Second, an apostrophe denotes tonal split induced by the aspiration of initial consonant [6, pp. 26-28]. Third, the letters denote tonal split conditioned by (a) long vowel or (b) short vowel [6, pp. 26-28] [7, 8]. Please be noted that these rules are historically conditioned, rather than synchronically correlated. For instance, the historical Tone 7 is not synchronically correlated with voiceless initials and has split into four tones in Malipo under the condition of vowel length and initial aspiration, among which T7'a is further merged with T1'. However, this paper is concerned with synchronic tonal phonology in the Malipo variety, with an aim to explore acoustic manifestations of tonal contrasts.

2. Methods

Eight speakers, four male and four female, from Tongta Village, Mengdong Yao Ethnic Township, Malipo County, Wenshan Zhuang and Miao Autonomous Prefecture, Yunnan Province, China, provided the speech data. They are all native Landian Yao and refer to themselves as /ki:m²¹mun²¹/. 64 monosyllabic words were used as test words and two repetitions were recorded, resulting in a total of 1,152 samples. The 16-bit audio speech was recorded into a Lenovo laptop through a Samson C01U Pro external condenser microphone at a sampling rate of 44,100 Hertz.

The rhyme was annotated as the tone-bearing unit in Praat 6.4.01 [9]. 11 equidistant points of fundamental frequency (F0) data were extracted from the tone-bearing unit, and the duration of the tone-bearing unit was measured. The F0 data were normalized using Log-Z-score and duration data were normalized using Z-score.

Growth Curve Analysis (GCA), a method within Linear Mixed-Effects Models (LMM), was employed to compare similarities and differences between F0 points along with their temporal changes [10]. This method has proven effective in the studies on complex Chinese tonal systems [11, 12]. The *lme4* package [13] in *R* [14] and *R Studio* [15] was used to build LMM models, where LZ values of the transformed pitch (F0) served as the dependent variable, and time (normalized), tone, gender, speaker, vowel type and coda type served as independent variables. After multiple fittings by adding or removing independent variables, the following model was chosen as the most suitable statistical model for the current study:

$$\text{Pitch (LZ value)} \sim (\text{Time} + \text{Time}^2) * \text{Tone} + \text{Vowel type} + (1 | \text{Speaker}) \quad (1)$$

This model performed best in terms of Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), log-likelihood ratio (logLik), and deviance, with chi-square test results supporting its significant superiority over other models ($p < 0.05$). The model uses a quadratic orthogonal polynomial fit, with tonal and vowel types as fixed variables and speakers as random variables.

After modeling these tones with quadratic orthogonal polynomials through GCA, three coefficients related to the pitch curve can be obtained: the intercept represents the overall pitch level of each tonal category, the linear coefficient of time in fixed effects represents the slope, and the quadratic coefficient of time represents the curvature. Therefore, to compare the similarities and differences in F0 points among these tones, we compared the differences in these three coefficients across different tone categories by obtaining Estimated Marginal Means (EMM) implemented in R through the *emmeans* package [16].

In addition, Linear Mixed-Effects Models were used to model the durational data of tones, with the final model being:

$$\text{Duration (Z value)} \sim \text{Tone} + \text{Codas type} + (1 | \text{Speaker}) \quad (2)$$

3. Results

3.1. Pitch and Tonal Patterns

As transcribed in Table 1, the 13 tones in the Malipo variety distinguish in 5 contour patterns: level, rising, falling, rise-fall, and fall-rise. Similar contours were compared below to examine how they are further differentiated from each other.

3.1.1. Level Tones

The Malipo variety has five level tones: T3', T6, T7b, T7'b, and T8. Figure 1 shows the pitch curves of these level tones under two different y-axis scales. The left graph has a y-axis range defined by the maximum and minimum LZ values from all tones, representing the tonal range of the Malipo variety. And all the five level tones cluster in the mid region of the tonal space. The right graph magnifies the y-axis. And the distinctions between the five level tones are better manifested.

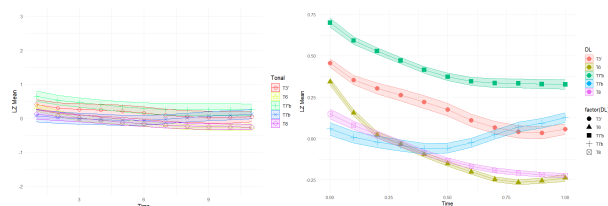


Figure 1: The five level tones in the Malipo variety. The X-axis represents normalized equal time intervals, and the Y-axis represents LZ values. Each pitch curve is composed of 11 pitch points, each point representing the mean LZ values of the speakers with the shaded area representing one standard deviation of the mean.

Figure 2 shows the EMM values of the five level tones in terms of intercept, slope, and quadratic following GCA in upper panel. And it also shows the results of Tukey's post-hoc tests in lower panel. First, two groups of tones, T7'b-T7b and T7'b-T8, distinguish in intercept. That is, the tones with the lowest intercept values, T7b and T8, differ significantly from the tone

with the highest intercept value, T7'b. Second, four groups of tones distinguish in slope, T3'-T7'b, T3'-T8, T6-T7'b and T6-T8. That is, tones with lower slope values, T6 and T3', differ significantly from those with higher slope values, T7'b and T8. Third, two groups of tones distinguish in quadratic, T3'-T7b and T6-T7b. That is, the tone with the lowest quadratic value, T7b, differs significantly from those with the highest values, T6 and T3'.

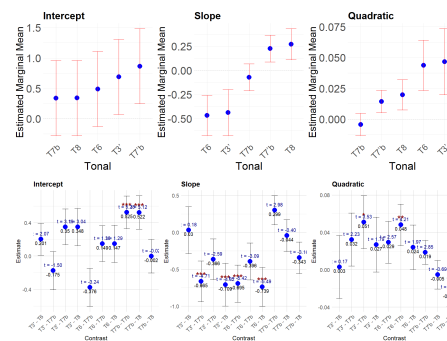


Figure 2: The EMM differences of the five level tones in the Malipo variety in terms of intercept, slope, and quadratic (upper), and the post-hoc tests on these coefficients (lower).

3.1.2. Falling Tones

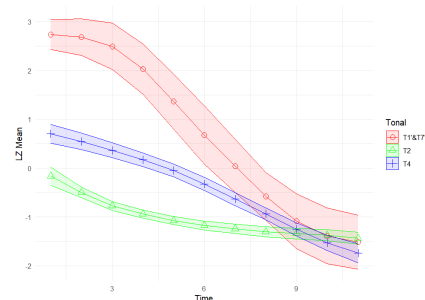


Figure 3: The three falling tones in the Malipo variety.

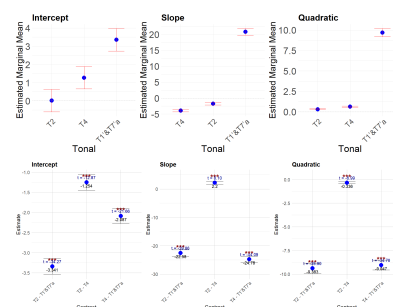


Figure 4: The EMM differences of the three falling tones in the Malipo variety in terms of intercept, slope, and quadratic (upper), and the post-hoc tests on these coefficients (lower).

There are three falling tones in the Malipo variety: T1'&T7'a, T2, and T4. Figure 3 shows the pitch curves of these falling tones, and Figure 4 shows the EMM values of these falling tones across different coefficients and the results of post-hoc tests. It can be observed that T1'&T7'a, T2, and T4 are

distributed from high, mid to low in the tonal space, making them easily distinguishable from one another. And the three falling tones show significant differences in terms of intercept, slope, and quadratic. However, the end points of these falling tones are at the same pitch level. That means that the differences in pitch contour, slope and curvature, could be attributed to the difference in initial pitch heights.

3.1.3. Rising Tones and Fall-Rise Tones

Figure 5 compares the rising tone T5, and the two fall-rise tones T1 and T7a together. The EMM results in Figure 6 show that T5 is significantly different from T1 and T7a in terms of intercept, suggesting that T5 has a higher pitch level, while T1 and T7a do not differ in pitch level. Nevertheless, the three tones differ significantly in slope and quadratic, and the differences between T5 and the other two tones are more pronounced. It is thus reasonable to treat T5 as a distinctive high rising tone. As for T1 and T7a, the two low concave tones manifest significant difference in contour pattern, namely 213 versus 212.

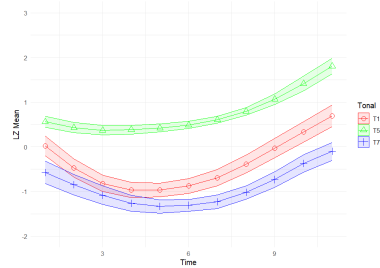


Figure 5: T5, T1, and T7a in the Malipo variety.

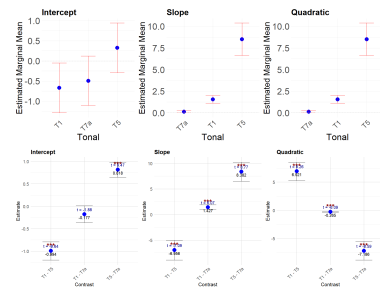


Figure 6: The EMM differences of T5, T1, and T7a in the Malipo variety in terms of intercept, slope, and quadratic (upper) and the post-hoc tests on these coefficients (lower).

3.1.4. Rise-Fall Tones

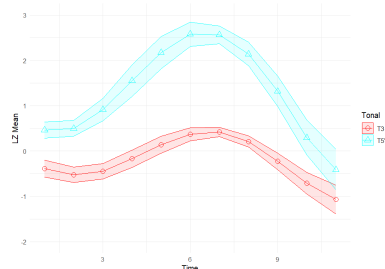


Figure 7: The rise-fall tones T3 and T5' in Malipo variety.

Figure 7 shows that the two rise-fall tones in the Malipo variety, T3 and T5', are located in the mid-high and mid-low regions of the tonal space, respectively. The EMM results in Figure 8 show that the two rise-fall tones have a significant difference in intercept, with T5' being higher than T3. But they do not have significant differences in slope and quadratic, indicating that they share a similar contour pattern of convexity.

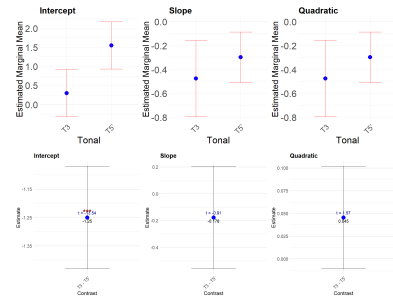


Figure 8: The EMM differences of the rise-fall tones in the Malipo variety in terms of intercept, slope, and quadratic (upper) and the post-hoc tests on these coefficients (lower).

3.2. Duration

Figure 9 shows that tones in checked syllables, T7a, T7b, T7c, and T8, have shorter durations than the other tones in syllables without a stop coda. After constructing LMM for all duration data, EMM values were obtained, and the results of post-hoc tests were shown in Figure 10.

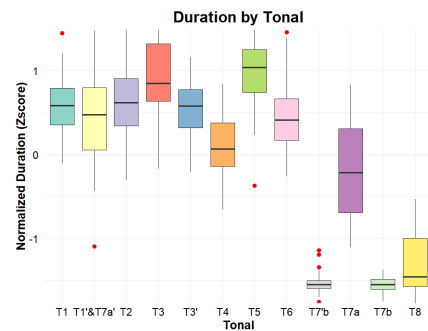


Figure 9: Duration of Tones in the Malipo variety.

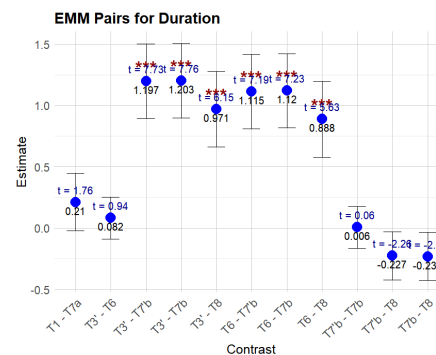


Figure 10: Results of Post-hoc Tests on EMM Values of Duration for Some Tones in Malipo variety.

Attention has been paid to the tones with similar contour shapes. First, it can be seen from the figure that there is no significant difference in duration between the two fall-rise tones T1 and T7a, suggesting that the only difference between them lies in concavity, with T1 having a sharper rising than T7a. As for the five level tones, comparisons reveal no significant difference with the unchecked (T3' and T6) and checked (T7b, T7'b, and T8) groups. But there is a significant difference in duration between the unchecked and checked groups. And the codas type yields a main effect: $F(3,457) = 16.905$, $p < 0.001$. That is, even though they have same pitch levels, unchecked and checked tones distinguish in duration

4. Conclusions

The Malipo variety has an extremely complex tonal system with 13 tones. The application of GCA enables us to examine tonal complexity from three perspectives: pitch, slope, and curvature. GCA successfully characterizes the 5 level tones, 3 falling tones, 3 rising or concave tones, and 2 convex tones in the Malipo variety, even though the differences between some tones are subtle. The most intriguing cases concern with the differences between the two concave tones and among the five level tones. The two concave tones differ in concavity only, and the five level tones first differ in duration and then further differ in pitch and/or contour within unchecked and checked groups respectively.

It should be admitted that the current study is solely based on F0 and durational data of tones. Further research is necessary to explore other aspects on tones. Complex tonal systems may rely on acoustic dimensions beyond F0 [17]. Therefore, it would be beneficial to examine whether phonation plays a role in differentiating tones, particularly those with subtle F0 differences. In addition to acoustic studies, perception studies are also required to determine whether the tones with subtle acoustic differences show a tendency to merge.

5. References

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